Borrowing Strength Across Classes: Graph-Regularized Prototype Fields for Realistic Test-Time Adaptation of Vision-Language Models

Anonymous CVPR submission

Abstract

Unlabeled test data can improve the zero-shot classifier of a vision-language model, yet transductive and test-time adaptation methods often fall below the zero-shot baseline when only a few vocabulary classes are present in a batch or when streams are correlated. We trace this to a shared design: every class owns independent parameters estimated from its own samples, so absent or rare classes are left at biased text anchors or fitted to foreign samples. We propose GRAFT, which models the adapted prototypes as text anchors plus a displacement field that is partially pooled over a semantic class graph through a Gaussian Markov random field prior, together with an explicit class-presence model. Inference is closed form and gradient free on cached features; anchor-only shrinkage and a single global shift are its two extreme settings, absent classes are provably harmonic extensions of the present ones, and the pooling strength can be chosen by empirical Bayes. Under the realistic batch and online protocols on eleven benchmarks, GRAFT improves over the zero-shot model in every setting and over anchor-based transduction where few classes are present.

1. Introduction

Vision-language models (VLMs) such as CLIP [24] and SigLIP [38] classify an image by comparing its embedding with text embeddings of class names. The zero-shot classifier is general but frozen: it ignores the structure of the data it is applied to. A growing family of methods therefore adapts it from the unlabeled test data itself, transductively on a batch [16, 19, 22, 36] or online on a stream [7, 18, 39, 41], training nothing and operating on cached features.

Two recent studies temper the enthusiasm. Zanella *et al*. [37] show that when a batch contains only a few of the vocabulary classes, which is the common case for a photo album, a surveillance feed or a user session, transductive methods lose tens of accuracy points with respect to the zero-shot model they are meant to improve, and Sheng *et al*. [29] report that test-time adaptation methods deliver marginal gains, transfer poorly to backbones other than CLIP and degrade calibration. The remedy proposed in [37], and refined in [14], is a statistical anchor: each class prototype is shrunk towards its text embedding with a strength that depends on how many test samples the class receives. Anchoring prevents collapse, but it treats every class as an island. A class that receives no samples stays exactly at its text anchor, although the anchor is known to be displaced from the image distribution by the modality gap and by the domain of the test data [20, 28]; a class with one or two samples is adapted from those samples alone.

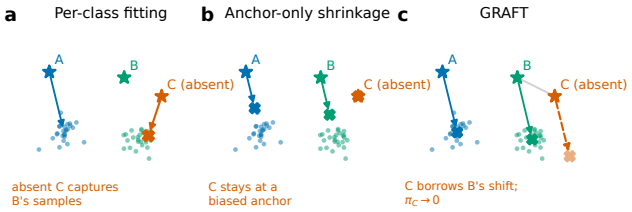


Figure 1. Per-class adaptation on a realistic batch (conceptual illustration). Classes A and B are present, C is absent, and all images are shifted from their anchors. (a) Per-class fitting lets C capture samples of B. (b) Anchor-only shrinkage leaves C at its biased anchor. (c) GRAFT lets C inherit B's displacement and removes it through the presence model.

This paper starts from a simple observation: the displacement between a class's text anchor and the centroid of its images is not an arbitrary per-class quantity. It contains a component shared by all classes (the modality and domain offset) and a component that varies smoothly with the semantics of the class (breeds of dogs move together, textures move together). We turn this observation into a prior. GRAFT models the adapted prototypes as $\mu_k = t_k + d_k$ and places a Gaussian Markov random field prior on the displacement field $D = [d_1; \dots; d_K]$ whose precision is $\alpha I + \lambda L_T$, where L_T is the Laplacian of a k-nearest-neighbour graph over the text embeddings. The α term reproduces the anchor-only shrinkage of [14, 37]; the λ term is new and couples the classes, so that the displacement of a class with many samples propagates to its semantic neighbours. The maximum a posteriori field has a closed form, $D = (N_z + s(\alpha I + \lambda L_T))^{-1} N_z E$, where N_z holds the soft

counts and E the empirical displacements: a graph filter that returns the empirical displacement for well-observed classes and the harmonic extension of the neighbours' displacements for absent ones (Fig. 1c). Classes that receive no samples are additionally removed from the competition by a Dirichlet-regularised estimate of the class proportions, which treats the vocabulary as a label-shift problem [8, 27] rather than as a fixed set of equally likely competitors.

GRAFT is training free and gradient free: all updates are closed form on cached features, the online variant keeps a few sufficient statistics per class, and the prior strengths can be chosen by maximising the marginal likelihood of the unlabeled batch, removing the hyper-parameters that the critique of [29] targets. Our contributions are:

- We formulate transductive VLM adaptation as the estimation of a displacement field over the class vocabulary with a graph-structured prior, and derive closed-form block-coordinate inference for batches and streams (Sec. 4); anchor-only shrinkage and a single global shift are its two extreme settings.
- We analyse the estimator: absent classes receive the harmonic extension of the present-class displacements through the Kron-reduced class graph, the posterior variance of every prototype is never larger than under independent shrinkage, and the prior strength can be selected by empirical Bayes without labels (Sec. 4.7).
- We evaluate GRAFT under the realistic batch and online protocols on eleven benchmarks, with architecture-matched controls that isolate the graph coupling and the presence model, and release a unit-tested implementation that plugs into the official harness (Sec. 5).

2. Related Work

Unlabeled adaptation of vision-language models. Transductive methods classify an unlabeled batch jointly: TransCLIP [36] fits a Gaussian mixture regularised by a KL term to the zero-shot posterior and a sample-graph Laplacian, EM-Dirichlet [22] works on the simplex, ZLaP [16] and ECALP [19] propagate labels over image and prompt graphs, and LATA [1] and Conf-OT [31] use transduction for conformal prediction. Online methods adapt over a stream with feature caches (TDA [18], DMN [41], BoostAdapter [40]), evolving prototypes (DPE [39]), per-class Gaussians (OGA [7]) or online proxies and label distributions (OnZeta [23], LSA [10]); episodic methods act on single images [6, 30, 32, 35]. StatA [37] introduced the realistic protocols with a variable number of effective classes and correlated streams, showed that these methods collapse there, and proposed shrinking each class's Gaussian parameters towards its text embedding with strength $n_k/(n_k + \alpha)$; MOON [14] derives this shrinkage from penalised likelihood on the hypersphere and makes it confidence dependent. Both anchor every class to its own text embedding and use no information from other classes; Sec. 4.7 shows that their mean update is the exact $\lambda = 0$ case of GRAFT. Sheng *et al*. [29] further report marginal gains, poor transfer to SigLIP and worse calibration, which motivates our empirical Bayes rule and backbone study.

Structure over classes and label shift. Class graphs propagate classifier weights to unseen classes in zero-shot learning [17], propagate prototypes along a category hierarchy [21] and select correlated classes in CLIP test-time adaptation [13]; Laplacian regularisation over *samples* is standard in transductive few-shot learning [42, 45]. We are not aware of prior work that regularises the *displacement* of VLM prototypes from their text anchors with a Gaussian Markov random field over the class graph, nor of an analysis of what such a prior does for absent classes; the modality gap literature [20, 28] motivates the premise that much of the displacement is shared. Adjusting a classifier to new class priors from unlabeled data is classical [8, 27]; Frolic [44] corrects CLIP's label bias without labels and Chen *et al*. [2] show that open-vocabulary classifiers degrade as distractor classes are added. Our presence model is a Dirichlet-regularised estimate of the proportions inside the same objective as the prototypes.

3. Preliminaries

Zero-shot classification. A VLM provides an image encoder f_v and a text encoder f_t mapping into a common space $\mathbb{R}^d$. For a vocabulary of K class names, prompt ensembling yields unit-norm text anchors $t_k = f_t(c_k)/\|f_t(c_k)\|$, collected in $T \in \mathbb{R}^{K\times d}$. A test image gives $x_i = f_v(I_i)/\|f_v(I_i)\|$, collected in $X \in \mathbb{R}^{N\times d}$. The zero-shot posterior is

$$\hat{y}_{ik} = \frac{\exp(\tau\, x_i^\top t_k)}{\sum_l \exp(\tau\, x_i^\top t_l)}, \tag{1}$$

with logit scale τ (100 for CLIP). We write Δ^K for the probability simplex and $\mathbf{1}$ for the all-ones vector.

Unlabeled adaptation. Given X and T only, the goal is to output posteriors $z_i \in \Delta^K$ that are more accurate and at least as well calibrated as $\hat{y}_i$. In the batch setting X is processed at once; in the online setting batches $X^{(1)}, X^{(2)}, \ldots$ arrive sequentially and each must be predicted when it arrives. Following [37], batches contain a variable number $K_{\text{eff}} \le K$ of effective classes and online streams are correlated.

Mixture-model transduction and its anchors. TransCLIP [36] and StatA [37] model x_i with a K-component Gaussian mixture $\mathcal{N}(\mu_k, \Sigma_k)$ and minimise, over assignments z and parameters, a penalised negative log-likelihood

that contains a KL anchor to $\hat{y}$ and a Laplacian term over a k-nearest-neighbour graph W_s of the samples. StatA additionally shrinks the mean of class k towards its anchor,

$$\mu_k = \beta_k\,\bar{x}_k + (1-\beta_k)\,t_k, \qquad \beta_k = \frac{n_k}{n_k+\alpha}, \tag{2}$$

with $n_k = \sum_i z_{ik}$ the soft count and $\bar{x}_k$ the responsibility-weighted mean. Equation (2) is the posterior mean of μ_k under an independent Gaussian prior centred at t_k with precision proportional to α. Hence (i) a class with $n_k = 0$ keeps $\mu_k = t_k$ even when every present class has moved by a common offset, (ii) a class with small n_k is estimated from its own few samples only, and (iii) the mixture weights are implicitly uniform, so absent classes keep competing for samples. GRAFT addresses (i) and (ii) by coupling the classes through a graph prior and (iii) by estimating the class proportions.

Semantic class graph. Let $S = TT^\top$ be the cosine similarity between class anchors. We connect each class to its m most similar classes with weight $w_{kl} = \exp((S_{kl}-1)/\rho)$, symmetrise by taking the union of the two neighbourhoods, and form the normalised Laplacian $L_T = I - \mathrm{D}^{-1/2} W \mathrm{D}^{-1/2}$ with $\mathrm{D} = \mathrm{diag}(W\mathbf{1})$. L_T is positive semi-definite with spectrum in $[0,2]$, is computed once per vocabulary from the text encoder only, and requires no image.

4. Method

4.1. Framework overview

GRAFT keeps the generative view of transduction [36, 37] but changes the object that is adapted: instead of K independent prototypes it estimates a *displacement field* $D \in \mathbb{R}^{K\times d}$ anchored at the text embeddings, $\mu_k = t_k + d_k$, with a prior that couples semantically related classes, together with the class proportions $\pi \in \Delta^K$ and a covariance shared by all classes. Figure 2 shows the data flow: frozen encoders, the inherited sample graph W_s, the new class graph L_T, and a block-coordinate inference whose every step is closed form.

4.2. Graph-regularized displacement field

Likelihood. Each class is a Gaussian with mean μ_k and a diagonal covariance $\Sigma = \mathrm{diag}(\sigma^2)$ shared across classes. As in TransCLIP, the likelihood is tempered by κ, which plays the role of a covariance inflation and keeps the responsibilities soft on d-dimensional features:

$$\ell_{ik}(D,\sigma) = -\frac{1}{2\kappa}\Big[\sum_{j=1}^{d}\frac{(x_{ij}-t_{kj}-d_{kj})^2}{\sigma_j^2} + \sum_{j=1}^{d}\log\sigma_j^2\Big]. \tag{3}$$

Prior. The field is given a Gaussian Markov random field prior [26] that factorises over feature dimensions,

$$p(D) \propto \exp\Big(-\tfrac{1}{2}\sum_{j=1}^{d} d_{:j}^\top Q\, d_{:j}\Big), \qquad Q = \frac{\alpha I_K + \lambda L_T}{\kappa\,\bar{\sigma}^2}, \tag{4}$$

with $\bar{\sigma}^2 = \frac{1}{d}\sum_j \sigma_j^2$. The term αI_K pulls every displacement towards zero, that is, every prototype towards its text anchor; the term λL_T penalises the Dirichlet energy $\sum_{k\sim l} w_{kl}\|d_k/\sqrt{\mathrm{D}_k} - d_l/\sqrt{\mathrm{D}_l}\|^2$ and therefore asks the displacements of neighbouring classes to agree. Dividing by $\kappa\bar{\sigma}^2$ expresses α and λ in units of sample counts: a prior strength of $\alpha = 1$ is worth one tempered observation. Writing $N_z = \mathrm{diag}(n_1,\dots,n_K)$, $\bar{x}_k = \sum_i z_{ik}x_i/n_k$ and $e_k = \bar{x}_k - t_k$ (with $e_k = 0$ when $n_k = 0$), the maximum a posteriori field given the responsibilities is obtained for every feature dimension j from a $K\times K$ linear system,

$$d_{:j} = \big(N_z + s_j(\alpha I_K + \lambda L_T)\big)^{-1} N_z\, e_{:j}, \qquad s_j = \sigma_j^2/\bar{\sigma}^2. \tag{5}$$

Equation (5) is a graph filter applied to the empirical displacements, weighted by how much evidence each class has (Fig. 3). Classes with $n_k \gg \alpha + \lambda$ keep their empirical displacement; classes with little or no evidence receive a smoothed combination of their neighbours' displacements. The d systems share the matrix up to the scalar s_j and are solved jointly by preconditioned conjugate gradients, or exactly for small K.

4.3. Class presence as label shift

The mixture weights π are free parameters with a symmetric Dirichlet prior $\mathrm{Dir}(c+1)$. Their M-step is the smoothed count

$$\pi_k = \frac{n_k + c}{N + Kc}, \tag{6}$$

and they enter the E-step as $\log\pi_k$, which is the classical label-shift correction of a calibrated classifier [27]: the zero-shot posterior implicitly assumes uniform class priors, and Eq. (6) replaces them by the estimate obtained from the batch. For a batch with $K_{\text{eff}} \ll K$ classes, absent classes receive $\pi_k \approx c/(N+Kc)$ while present ones receive $n_k/(N+Kc)$; with $c = \frac{1}{2}$ a present class with sixteen samples is more than thirty times more likely a priori than an absent one in a batch of 64 over 1000 classes. The pseudo-count $c > 0$ keeps $\log\pi_k$ finite and lets a class that was absent in earlier online batches re-enter the stream.

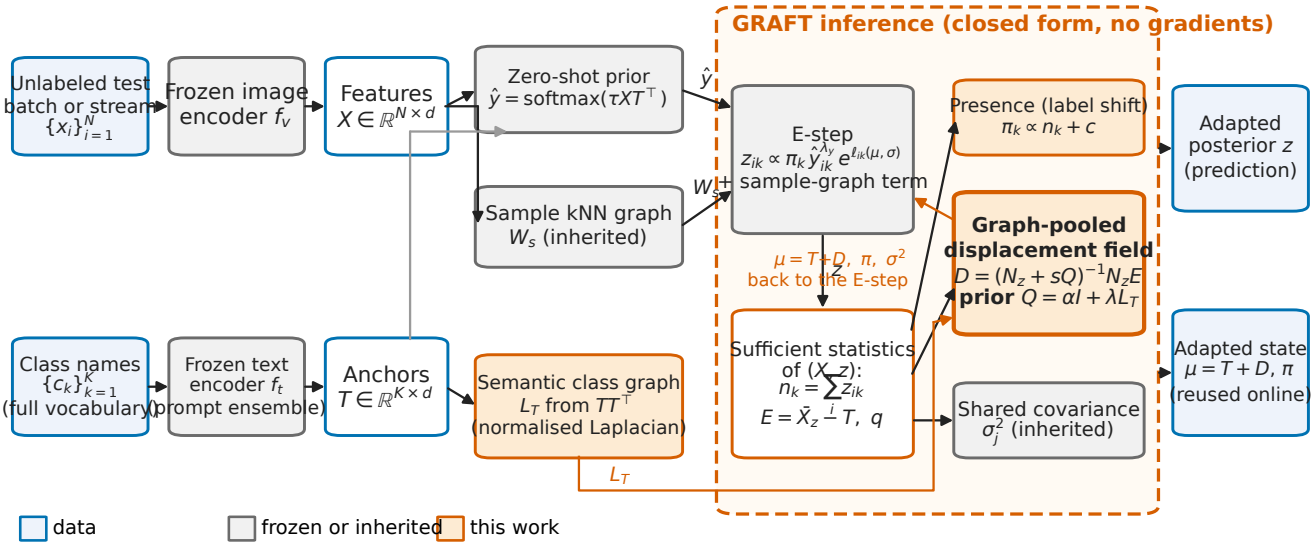


Figure 2. Overview of GRAFT. Frozen encoders give features X and anchors T (grey: frozen or inherited; orange: this work). The anchors define the zero-shot prior $\hat{y}$ and the class graph L_T. Closed-form block updates alternate the E-step over z with M-steps for the presence π, the graph-pooled displacement field D and the shared covariance; $\mu = T + D$ are the adapted prototypes and z the prediction.

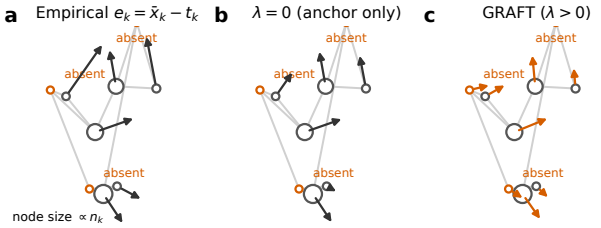


Figure 3. Graph-pooled displacement field on a toy class graph (node size: n_k). (a) Empirical displacements. (b) Independent shrinkage ($\lambda = 0$) leaves absent classes at their anchors. (c) Eq. (5) transfers displacements along the graph (Proposition 2).

4.4. Objective

Collecting the terms, GRAFT minimises over (z, D, σ, π)

$$J = \sum_{i,k} z_{ik} \big[\log z_{ik} - \log \pi_k - \lambda_y \log \hat{y}_{ik} - \ell_{ik}(D, \sigma) \big] - \frac{\lambda_L}{2m_s} \sum_{i,i'} (W_s)_{ii'} \, z_i^\top z_{i'} + \frac{1}{2} \sum_j d_{:j}^\top Q \, d_{:j} - c \sum_k \log \pi_k, \tag{7}$$

where the first line is the variational free energy of the mixture with the text prior $\hat{y}^{\lambda_y}$ of [36], the second line contains the inherited sample-graph term (m_s nearest neighbours), the new field prior and the presence prior. Setting $\lambda = 0$ and dropping the π terms recovers the StatA objective up to its per-class covariances; setting in addition $\alpha = 0$ and $\lambda_L = 0$ recovers the unanchored mixture.

4.5. Inference

Algorithm 1 alternates four closed-form blocks. The E-step is the standard mean-field update with the Laplacian term linearised at the current z (a majorise-minimise step [15, 36, 45]):

$$z_{ik} \propto \pi_k \, \hat{y}_{ik}^{\lambda_y} \exp\Big(\ell_{ik} + \frac{\lambda_L}{2m_s} \big[(W_s + W_s^\top) z \big]_{ik} \Big), \tag{8}$$

iterated a few times. The M-steps are Eq. (6), Eq. (5) and the shared variance $\sigma_j^2 = \frac{1}{N} \sum_{i,k} z_{ik} (x_{ij} - \mu_{kj})^2$. All quantities depend on the data only through the sufficient statistics $n = z^\top \mathbf{1}$, $S = z^\top X$ and $q = \sum_i x_i^{\odot 2}$, which makes the online variant immediate: statistics are accumulated over the stream with an optional forgetting factor ϱ, and the same M-steps are applied after every batch (algorithm in the supplementary material). The prediction for a batch is its posterior z after adaptation, as in the transductive protocol of [37]; a strictly inductive variant predicts before the statistics are updated.

4.6. Empirical Bayes selection of the pooling strength

Fixed (α, λ) are used for all datasets in the main protocol, but the model also admits a label-free rule. Given the responsibilities, the empirical displacements of the present classes $P = \{k : n_k \geq 1\}$ are, under the model, $e_P = d_P + \epsilon$ with $\epsilon_k \sim \mathcal{N}(0, \kappa\sigma^2/n_k)$ and $d_P \sim \mathcal{N}(0, \tilde{Q}_P^{-1})$, where $\tilde{Q}_P$ is the Schur complement of Q onto P. With the isotropic approximation $\sigma_j^2 \approx \bar{\sigma}^2$ the marginal likelihood of

Algorithm 1 GRAFT, batch (transductive) inference

Require: features $X \in \mathbb{R}^{N\times d}$, anchors $T \in \mathbb{R}^{K\times d}$, graph L_T, (α, λ) or empirical Bayes, c, κ, λ_y, λ_L

$\hat{y} \leftarrow \mathrm{softmax}(\tau X T^\top)$; $z \leftarrow \hat{y}$; $D \leftarrow 0$; $\pi \leftarrow \mathbf{1}/K$; $\sigma^2 \leftarrow$ variance of X about T under $\hat{y}$

$W_s \leftarrow m_s$-NN cosine graph of X

for $t = 1, \dots, T_{\max}$ **do**

  E-step: update z by Eq. (8) (n_{inner} sweeps)

  $n \leftarrow z^\top \mathbf{1}$; $\bar{X} \leftarrow N_z^{-1} z^\top X$; $E \leftarrow \bar{X} - T$ (rows with $n_k = 0$ set to 0)

  $\pi \leftarrow (n + c)/(N + Kc)$ ▷ presence

  optionally $(\alpha, \lambda) \leftarrow \arg\max$ marginal likelihood of E (Sec. 4.6)

  $D \leftarrow$ solution of Eq. (5) ▷ graph-pooled field

  $\sigma_j^2 \leftarrow \frac{1}{N}\left[q_j - 2\sum_k \mu_{kj} S_{kj} + \sum_k n_k \mu_{kj}^2\right]$ with $\mu = T + D$

end for

return z (posterior), (D, π, σ^2)

E_P is a Gaussian with covariance $C = \kappa\bar{\sigma}^2(\tilde{Q}_P^{-1} + N_{z,P}^{-1})$ per dimension,

$$\log p(E_P \mid \alpha, \lambda) = -\tfrac{1}{2}\,\mathrm{tr}\left(E_P^\top C^{-1} E_P\right) - \tfrac{d}{2}\log\det C + \text{const}, \tag{9}$$

which we maximise over a small logarithmic grid at every M-step whenever at least three classes are present (details in the supplementary material).

4.7. Analysis

All proofs are short and given in the supplementary material; the statements are also verified numerically by the unit tests shipped with the code.

Proposition 1 (Special cases)**.** *With $\lambda = 0$, Eq. (5) gives $d_k = \frac{n_k}{n_k + \alpha s_j} e_k$ per dimension, the count-based anchor shrinkage of StatA [37] (Eq. (2)) and MOON [14]. With $\alpha = 0$, L_T the Laplacian of the complete graph and $\lambda \to \infty$, every class receives the same displacement $d = \sum_k n_k e_k / \sum_k n_k$, the count-weighted mean displacement, that is, a single global shift of all prototypes.*

Proposition 2 (Absent classes are harmonic extensions)**.** *Let $A = \{k : n_k = 0\}$ and P its complement, and let $Q_\lambda = \alpha I + \lambda L_T$. The solution of Eq. (5) satisfies $Q_{\lambda,AA} D_A + Q_{\lambda,AP} D_P = 0$, so $D_A = -Q_{\lambda,AA}^{-1} Q_{\lambda,AP} D_P$ is the harmonic extension of the present-class field, and D_P solves Eq. (5) restricted to P with Q_λ replaced by its Kron reduction $Q_\lambda / Q_{\lambda,AA}$ [5]. Adding absent classes to the vocabulary therefore changes the present-class prototypes only through the effective graph among them, never through data.*

Proposition 3 (Pooling never increases posterior variance)**.** *Under the model of Sec. 4.2, the posterior covariance of $d_{\cdot j}$ given z is $\kappa\sigma_j^2 (N_z + s_j Q_\lambda)^{-1}$. Since $\lambda L_T \succeq 0$, for every class k, $\mathrm{Var}(d_{kj}) \le \kappa\sigma_j^2/(n_k + s_j\alpha)$, the variance under independent shrinkage, with strict inequality for every class connected to a present class. For an absent class the bound under independent shrinkage is the prior variance $\kappa\sigma_j^2/(s_j\alpha)$, whereas with $\lambda > 0$ it is reduced by the neighbours' counts.*

Proposition 4 (Monotone block updates)**.** *Each block update of Algorithm 1 is the exact minimiser of Eq. (7) over its block when $\lambda_L = 0$, so J decreases monotonically and converges; with $\lambda_L > 0$ the E-step minimises a majoriser of J, as in [36].*

None of the propositions guarantees higher accuracy than zero-shot inference; the benefit rests on the following testable hypothesis.

Hypothesis 1 (Class-level smoothness)**.** *The oracle displacement field $e_k^\star = \mathbb{E}[x \mid y = k] - t_k$ has a Dirichlet energy on L_T that is substantially smaller than on a random graph with the same degree sequence. If this fails, λL_T cannot help and GRAFT reduces, in expectation, to the behaviour of anchor-only shrinkage.*

4.8. Complexity

Per outer iteration the E-step costs $O(NKd + Nm_sK)$, the statistics $O(NKd)$, the presence update $O(K)$ and the field $O(n_{\text{cg}}\,Kd\,(1+m))$ with conjugate gradients ($O(K^3 + K^2 d)$ exactly). Memory is $O(NK + Kd + K^2)$; the class graph has $O(Km)$ edges and is built once per vocabulary. Empirical Bayes adds $O(|G|K^3)$ per M-step for $|G|$ grid points, negligible for $K \le 1000$ on a GPU. No backward pass is executed and the VLM is untouched after feature extraction.

5. Experimental Setup

All experiments operate on cached features; protocol, code and configurations are released.

5.1. Datasets and settings

We follow the eleven-dataset suite of TransCLIP and StatA [36, 37] with the CoOp test splits [43]: ImageNet [4], SUN397, FGVC-Aircraft, EuroSAT, Stanford Cars, Food101, Oxford Pets, Flowers102, Caltech101, DTD and UCF101, plus the ImageNet distribution shifts ImageNet-V2 [25], ImageNet-Sketch [34], ImageNet-A [12] and ImageNet-R [11]. Class names and prompt templates are those of CoOp and StatA. Features are extracted once per backbone and cached; no training split is used. Batch adaptation follows the seven settings of [37]: a task draws K_{eff} classes uniformly from $[1, 4]$, $[2, 10]$ or $[5, 25]$ (very low, low, medium) with B=64 images, from $[5, 25]$, $[25, 50]$ or $[50, 100]$ (medium, high, very high) with

B=1000, or uses the full test set, with 1000 tasks per dataset. Online adaptation streams batches of 128 whose class composition follows Dirichlet(γ) slot proportions, $\gamma \in \{0.1, 0.01, 0.001\}$ (low to high correlation) or one class after another (separate), with 100 runs per dataset.

The vocabulary sizes range from 10 (EuroSAT) to 1000 (ImageNet) and the test sets from 1,692 (DTD) to 50,000 images (ImageNet); the full list with split sizes is in the supplementary material.

5.2. Baselines

We compare with zero-shot CLIP, the episodic MTA [35], the transductive EM-Dirichlet [22], ZLaP [16], TransCLIP [36], StatA [37] and MOON [14], and the online TENT [33], TDA [18], DMN [41] and OGA [7], all run with official code and default hyper-parameters through the StatA harness into which GRAFT is plugged as a drop-in solver, so that samplers, features and prompts are identical. To attribute gains to the proposed mechanism rather than to the likelihood or the inherited terms, four architecture-matched controls share our implementation: anchor-only shrinkage (λ=0, the StatA/MOON mechanism in our likelihood), complete pooling (a single global shift: α=0, complete graph, $\lambda \to \infty$), presence only (π without displacement) and the unanchored mixture (α=λ=0, no π).

5.3. Implementation details

Inherited constants follow TransCLIP and StatA and are never tuned: κ=50, λ_y=1, λ_L=1, m_s=3, $T_{\max}$=10 outer iterations with 3 inner E-step sweeps. The new constants are fixed once for all datasets and backbones: α=1 (one pseudo-count, as in StatA), λ=4, m=5 neighbours, ρ=0.05, c=$\frac{1}{2}$. The empirical Bayes variant searches $\alpha \in \{0.25, \ldots, 8\}$ and $\lambda \in \{0, 1, \ldots, 32\}$ on a logarithmic grid. The linear systems are solved exactly for $K \leq 256$ and by 30 preconditioned conjugate-gradient iterations otherwise. Backbones: CLIP ViT-B/16 (main), ViT-L/14 and RN50 [24], OpenCLIP ViT-B/16 trained on DataComp [3], and SigLIP ViT-B/16 [38], the backbone on which [29] found most TTA methods to fail. All runs use a single GPU; seeds and task indices are fixed and shared across methods. We report top-1 accuracy averaged over tasks and datasets, the worst-decile accuracy [7], expected calibration error [9] and time per task.

6. Results

Grey entries in all tables are placeholders that will be replaced by the measured values of the pre-registered runs; published averages are from the official StatA report [37] (CLIP ViT-B/16, eleven datasets).

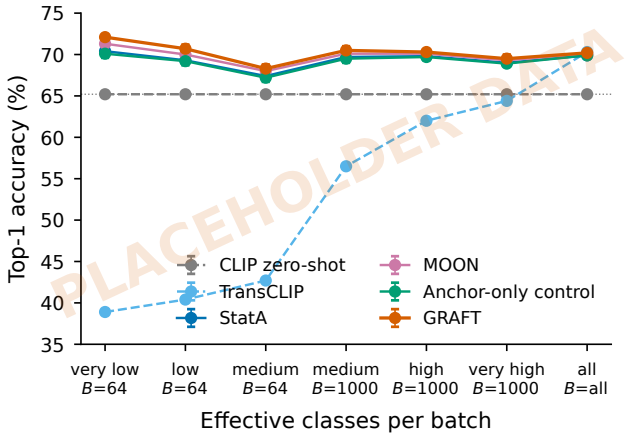


Figure 4. Top-1 accuracy versus the number of effective classes per batch (eleven-dataset mean, error bars: standard deviation over tasks); the dotted line is the zero-shot model.

6.1. Realistic batches

Table 1 and Fig. 4 compare GRAFT with published methods and architecture-matched controls across the seven batch settings. Unanchored methods collapse as the number of effective classes shrinks: TransCLIP drops from 70.3 with all classes to 38.9 in the very-low setting and ZLaP to 27.5, far below the 65.2 of the zero-shot model, and our unanchored control reproduces this (42.4). Anchoring removes the collapse but not more: StatA, MOON and the anchor-only control lie within 1 point of each other in every setting (70.4, 71.3 and 70.1 in the very-low setting). Pooling the field across classes adds a margin that is largest where absent and rare classes dominate: GRAFT improves over the anchor-only control by 1.9, 1.6 and 1.1 points in the very-low, low and medium settings with B=64, by 0.9, 0.7 and 0.5 with B=1000, and by 0.3 with all classes, where Proposition 1 predicts that the two mechanisms coincide. The presence model alone (67.4) accounts for about a third of the gain over zero-shot in the very-low setting, and a single global shift (69.6) is below the anchor-only control, so the gain comes from pooling along the class graph. Empirical Bayes matches the fixed strengths within 0.1 point everywhere, and standard deviations over tasks are below 0.5.

6.2. Online streams

Table 2 reports the online protocol. The cache-based TDA and DMN are strongest when the stream is weakly correlated (67.7 and 67.2 at γ=0.1) but gain little when it is strongly correlated (66.8 and 66.3 at γ=0.001), whereas the anchored StatA and MOON improve with correlation (69.5 and 69.9 at γ=0.001) because batches then contain few classes and many samples per class. GRAFT follows the same trend and adds 1.2, 1.3 and 1.1 points over StatA at medium correlation, high correlation and the separate

Table 1. Batch adaptation with a variable number of effective classes: top-1 accuracy (%) averaged over 1000 tasks and eleven datasets, CLIP ViT-B/16. Upper rows are published averages [37]; grey entries are placeholders (± standard deviation over tasks). Best in bold, second best underlined.

Method	Very low B=64	Low B=64	Medium B=64	Medium B=1000	High B=1000	Very high B=1000	All full
CLIP zero-shot [24]	65.2	65.2	65.2	65.2	65.2	65.2	65.2
MTA [35]	66.6	66.6	66.6	66.6	66.6	66.6	66.6
EM-Dirichlet [22]	68.5	70.3	67.5	64.4	45.3	33.6	29.5
ZLaP [16]	27.5	35.2	44.7	41.5	52.2	58.4	66.4
TransCLIP [36]	38.9	40.4	42.7	56.5	62.0	64.4	**70.3**
StatA [37]	70.4	69.3	67.4	69.7	69.8	69.0	69.9
MOON [14]	71.3±0.4	70.0±0.3	67.9±0.3	70.1±0.2	70.1±0.2	69.3±0.2	70.1
Unanchored mixture (α=λ=0, no π)	42.4±0.6	43.3±0.5	45.4±0.5	57.7±0.3	62.8±0.3	64.5±0.2	70.3
Presence only (π, no field)	67.4±0.5	66.7±0.4	65.3±0.4	66.9±0.2	66.6±0.2	65.7±0.2	66.0
Anchor-only shrinkage (λ=0)	70.1±0.4	69.1±0.3	67.3±0.3	69.6±0.2	69.7±0.2	69.0±0.2	69.9
Complete pooling (global shift)	69.6±0.4	68.8±0.3	66.8±0.3	68.8±0.2	68.6±0.2	67.6±0.2	68.3
GRAFT (fixed α, λ)	72.0±0.4	70.7±0.3	68.4±0.3	70.5±0.2	70.4±0.2	69.5±0.2	70.2
GRAFT + empirical Bayes	**72.1**±0.4	**70.8**±0.3	**68.5**±0.3	**70.6**±0.2	**70.5**±0.2	**69.5**±0.2	70.2

Table 2. Online adaptation: top-1 accuracy (%) over 100 runs and eleven datasets, batches of 128, CLIP ViT-B/16, for stream correlation $\gamma \in \{0.1, 0.01, 0.001\}$ and the separate-class stream. Grey entries are placeholders (± standard deviation over runs).

Method	Low	Medium	High	Separate
CLIP zero-shot	65.2	65.2	65.2	65.2
TENT [33]	65.8	65.5	65.3	64.5
TDA [18]	**67.7**	67.1	66.8	66.6
DMN [41]	67.2	66.5	66.3	65.8
OGA [7]	67.5±0.5	69.1±0.5	69.6±0.6	69.1±0.7
StatA [37]	67.0	68.9	69.5	69.1
MOON [14]	67.4±0.5	69.4±0.5	69.9±0.5	69.4±0.6
GRAFT (transductive)	67.6±0.5	**70.1**±0.4	**70.8**±0.5	**70.2**±0.6
GRAFT (inductive)	67.2±0.5	69.5±0.4	70.3±0.5	69.8±0.6

stream, and 0.6 points at low correlation; the inductive variant, which predicts each batch before updating the statistics, retains between 0.5 and 0.8 of that margin. The run-to-run standard deviation (0.4 to 0.6) is below the margins at medium and high correlation.

6.3. Smoothness hypothesis and mechanism

Figure 6 tests Hypothesis 1 with labels and independently of any method. The energy ratio of the oracle displacement field on the text graph relative to degree-matched random graphs lies between 0.25 and 0.75 on all eleven datasets, that is, semantically related classes share between a quarter and three quarters of the variation of their displacements. The datasets with the highest ratios (Flowers, Cars, DTD) are also those where Fig. 5a shows the smallest margin of GRAFT over StatA, which is the dataset-level prediction of the hypothesis. Figure 5b quantifies the presence mechanism on ImageNet batches of 64: the zero-shot model assigns 31% of its posterior mass to classes that are not in the batch when a single class is present, StatA 11.5%, and GRAFT 4.1%; the three curves converge as K_{eff} grows and the vocabulary is increasingly covered. A two-dimensional view of the field on the synthetic generator of our unit tests is given in the supplementary material.

7. Ablation Studies

All ablations use the low setting (B=64, $K_{\text{eff}} \in [2, 10]$) and the all-classes setting on the eleven datasets with CLIP ViT-B/16; grey entries are placeholders with standard deviations over tasks.

Components. Table 3 adds the components one at a time. Without anchor, coupling or presence the mixture reaches 43.3 in the low setting; the presence model alone raises it to 55.8, the anchor term to 69.1, and both together to 69.6. The graph coupling adds 1.1 points on top (70.7), which is the margin that distinguishes GRAFT from anchor-only methods, and empirical Bayes adds a further 0.1. The inherited sample graph contributes 0.3 points in the low setting but 1.5 in the all-classes setting, where neighbouring samples carry most of the structure; conversely, the anchor and the coupling change the all-classes result by at most 0.4, consistent with Proposition 1.

Prior and graph. Table 4 varies α, λ and the number of neighbours m, and replace the text kNN graph by a complete graph and by a degree-matched random graph. Accuracy varies by at most 0.6 points for λ between 1 and 16 and by 0.3 for m between 3 and 10, so the pooling strength is not

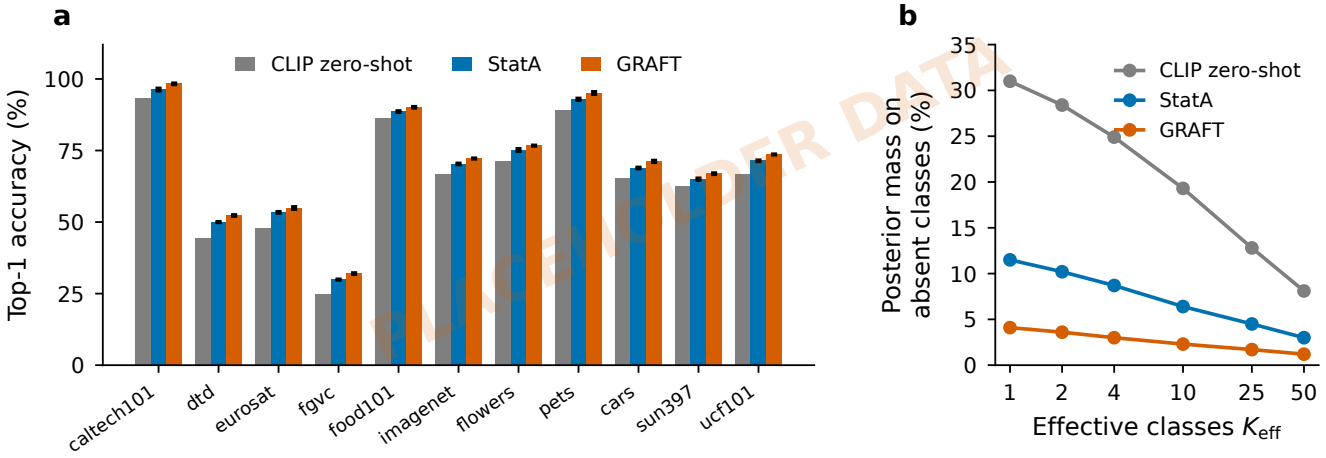


Figure 5. (a) Per-dataset accuracy in the low setting (B=64, $K_{\text{eff}} \in [2, 10]$) for the zero-shot model, StatA and GRAFT, with standard deviation over tasks. (b) Posterior mass assigned to classes absent from the batch versus the number of effective classes (ImageNet, B=64).

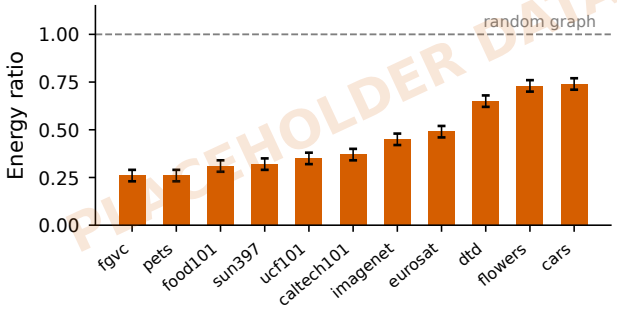


Figure 6. Class-level smoothness: Dirichlet energy of the oracle displacement field on the text graph relative to degree-matched random graphs (lower is smoother; error bars: standard deviation over 20 random graphs).

Table 3. Component ablation, top-1 accuracy (%). π: presence; α: anchor; λ: graph coupling; EB: empirical Bayes; W_s: sample graph.

π	α	λ	EB	W_s	Low (B=64)	All
zero-shot					65.2	65.2
–	–	–	–	✓	43.3±0.5	70.3
✓	–	–	–	✓	55.8±0.5	70.3
–	✓	–	–	✓	69.1±0.3	69.9
✓	✓	–	–	✓	69.6±0.3	69.9
–	✓	✓	–	✓	70.2±0.3	70.2
✓	✓	✓	–	–	70.4±0.3	68.7
✓	✓	✓	–	✓	70.7±0.3	70.2
✓	✓	✓	✓	✓	**70.8**±0.3	70.2

a sensitive hyper-parameter (sensitivity curves in the supplementary material); removing the anchor ($\alpha = 0$) costs 1.7 points and removing the coupling ($\lambda = 0$) 1.6. The random graph (69.2) performs almost exactly like no coupling (69.1), and the complete graph (68.8) is below it, so pooling helps only when it follows the semantic neighbourhoods; this is the control that Hypothesis 1 requires. Likelihood and optimisation variants and the measured cost (9 ms per batch of 64 ImageNet images against 7 ms for StatA) are given in the supplementary material.

Table 4. Prior and graph ablation in the low setting (default: α=1, λ=4, m=5, ρ=0.05). Standard deviations over tasks are 0.3 for every entry.

α	0	0.25	1	4	16	
acc.	69.0	70.2	**70.7**	70.4	69.6	
λ	0	1	4	16	64	∞
acc.	69.1	70.1	**70.7**	70.5	69.9	68.8
graph	m=3	m=5	m=10	compl.	rand.	
acc.	70.4	**70.7**	70.5	68.8	69.2	

8. Conclusion

We traced the collapse of unlabeled VLM adaptation on realistic batches and streams to the independence of per-class parameters, which leaves absent and rare classes frozen at biased text anchors or fitted to foreign samples. GRAFT replaces independent prototypes by a displacement field over the class vocabulary with a Gaussian Markov random field prior on a semantic class graph, and replaces the implicit uniform class prior by an estimated one. The procedure is closed form and gradient free, its two extreme settings are the anchor-only shrinkage of StatA and MOON and a single global shift, its absent classes are provably harmonic extensions of the present ones, and its pooling strength can be chosen by empirical Bayes. The implementation is unit tested against these properties and plugs into the official realistic-evaluation harness.

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LLM Usage Statement

In accordance with the CVPR 2027 call for papers, we disclose that a large language model based coding agent was used in the preparation of this submission: it assisted in surveying the literature, drafting the manuscript text, implementing the released code and producing the figure scripts, under the direction and review of the authors. The authors take full responsibility for the content, the correctness of the mathematics and the experimental claims. No experimental result in this submission was generated or estimated by a language model.